

Spearman correlation

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Spearman correlation measures the strength and direction of association between two ranked variables. It is a non-parametric measure, which means it does not assume a linear relationship or require the data to be normally distributed. Instead, it assesses how well the relationship between two variables can be described by a monotonic function.

0.0.1 Mathematical Explanation:

1. **Rank the Data:** Convert the data values of each variable to ranks.
2. **Difference of Ranks:** For each pair of ranks, compute the difference d_i between the two ranks.
3. **Square the Differences:** Compute d_i^2 for each pair of ranks.
4. **Sum of Squares:** Calculate the sum $\sum d_i^2$.
5. **Spearman's Correlation Coefficient ρ** is computed using the formula:

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

where n is the number of paired ranks.

0.0.2 Python Example:

Here's how to compute Spearman correlation using Python with the `scipy` library:

```
import numpy as np
from scipy.stats import spearmanr

# Sample data
x = [1, 2, 3, 4, 5]
y = [5, 6, 7, 8, 7]

# Compute Spearman correlation
correlation, p_value = spearmanr(x, y)

print(f"Spearman correlation: {correlation}, p-value: {p_value}")
```

0.0.3 Output:

This script will output the Spearman correlation coefficient and the p-value, indicating the strength and statistical significance of the correlation between the two variables.

1 For Images

Spearman correlation assesses the rank-order relationship between two variables. For images, this involves treating the pixel values as one-dimensional arrays and computing the Spearman correlation between them.

1.0.1 Steps:

1. **Flatten the Images:** Convert the images into 1D arrays.
2. **Rank the Values:** Assign ranks to the pixel values in both arrays.
3. **Calculate Spearman Correlation:** Use the formula:

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

where d_i is the difference between the ranks of corresponding pixels, and n is the number of pixels.

1.0.2 Python Example:

You can use libraries like `numpy` and `scipy` to calculate the Spearman correlation.

```
import numpy as np
from scipy.stats import spearmanr
from PIL import Image

# Load two images
image1 = Image.open('image1.png').convert('L') # Convert to grayscale
image2 = Image.open('image2.png').convert('L') # Convert to grayscale

# Flatten the images to 1D arrays
data1 = np.array(image1).flatten()
data2 = np.array(image2).flatten()

# Calculate Spearman correlation
correlation, _ = spearmanr(data1, data2)

print(f"Spearman correlation: {correlation}")
```

1.0.3 Explanation of Code:

- **Image Loading:** The images are loaded and converted to grayscale.
- **Flattening:** The pixel values are turned into 1D arrays using `.flatten()`.
- **Spearman Calculation:** The `spearmanr()` function from `scipy.stats` computes the correlation.

This method can effectively measure the monotonic relationship between the pixel intensity distributions of the two images.